

Ischemia prediction score (IsPS) in patients with strangulated small bowel obstruction: a retrospective cohort study.

Murao S, Fujino S, Danno K, Takeda T, Yamamoto K, Higashiguchi M et al.

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Backgrounds

Intestinal ischemia of strangulated small bowel obstruction (SSBO) requires prompt identification and early intervention. This study aimed to evaluate the risk factors and develop a prediction model of intestinal ischemia requiring bowel resection in SSBO.

Methods

This was a single-center, retrospective cohort study of consecutive patients underwent emergency surgery for SSBO from April 2007 to December 2021. Univariate analysis was performed to identify the risk factors for bowel resection in these patients. Two clinical scores (with contrasted computed tomography [CT] and without contrasted CT) were developed to predict intestinal ischemia. The scores were validated in an independent cohort.

Results

A total of 127 patients were included, 100 in the development cohort (DC) and 27 in the validation cohort (VC). Univariate analysis showed that high white blood cell count (WBC), low base excess (BE), ascites and reduced bowel enhancement were significantly associated with bowel resection. The ischemia prediction score (IsPS) comprised 1 point each for $WBC \geq 10,000/L$, $BE \leq -1.0 \text{ mmol/L}$, ascites, and 2 points for reduced bowel enhancement. The simple IsPS (s-IsPS, without contrasted CT) of 2 or more had a sensitivity of 69.4%, specificity of 65.4%. The modified IsPS (m-IsPS, with contrasted CT) of 3 or more had a sensitivity of 86.7%, specificity of 76.0%. AUC of s-IsPS was 0.716 in DC and 0.812 in VC, and AUC of m-IsPS was 0.838 and 0.814.

Conclusion

IsPS predicted possibility of ischemic intestinal resection with high accuracy and can help in the early identification of intestinal ischemia in SSBO.

Supplementary Information

The online version contains supplementary material available at [10.1186/s12876-023-02761-z](https://doi.org/10.1186/s12876-023-02761-z).

Keywords

Introduction

Strangulated small bowel obstruction (SSBO) is one of the leading causes of acute abdomen that requires an emergency operation and accounts for 20% of emergency surgical procedures [1]. When intestinal ischemia or necrosis occurs in SSBO, delayed diagnosis and intervention can result in a higher incidence of mortality, ranging from 8 to 25% [2]. However, the evaluation of intestinal ischemia is not well standardized, and most non-specialized clinicians are often unable to distinguish between a critical SSBO requiring bowel resection and others.

Several studies have reported the various clinical, laboratory and radiological risk factors for intestinal ischemia in SSBO patients. Clinical signs of intestinal ischemia include fever, pain duration, guarding, leucocytosis, peritoneal fluid, or reduced bowel contrast enhancement on computed tomography (CT) [3–5]. However, each clinical parameter alone was poorly predictive with a sensitivity of 48% in physical examination or a diminished specificity (50–64%) of bowel ischemia as a radiological sign [6–8]. Thus, there is no consensus on indications for surgical small bowel resection for specific SSBO patients.

Moreover, in most studies, radiological signs were based on the evaluation of experienced radiologists, which is not available in some clinical settings, and thus there is a need for objective radiological assessment with widely applicable criteria.

This study evaluated the risk factors for bowel resection in SSBO patients who underwent emergency laparotomy using objective radiological criteria for assessing CT. Thereafter, we developed and validated the prediction score for the need of bowel resection in SSBO.

Methods

Study design

This was a single-center, retrospective cohort study conducted at a tertiary hospital in Osaka, Japan. Records of patients who underwent operation after being diagnosed with strangulated small bowel obstruction between April 2007 and December 2021 were retrospectively reviewed. Exclusion criteria were large bowel obstruction, incarcerated abdominal wall hernia, femoral hernia and patients who received conservative treatment initially. The included patients were divided into 2 cohorts according to the operation period. The development cohort (DC) included patients treated between April 2007 and December 2018, and the validation cohort (VC) included patients treated between December 2018 and December 2021. The study followed the Declaration of Helsinki and was approved by the Institutional Review Board at Minoh City Hospital (R0311B64).

Data collection

All clinical and biological data were collected during the admission and included age, sex, past-history of surgery, American Society of Anesthesiologists physical status (ASA-PS), duration of symptoms before surgery, peritoneal irritation signs (guarding, rebound) and body temperature. Blood tests included white blood cell (WBC) count, level of C-reactive protein (CRP), lactate dehydrogenase (LDH), creatine kinase (CK) and blood gas (pH, PaCO₂ and Base Excess). CT findings included the presence of ascites, mesenteric fluid and reduced bowel wall enhancement. Reduced wall enhancement was evaluated as discussed here. The CT value was continuously measured from the inner side to the outer side of the distended small-bowel wall with a circle of 10 mm diameter as region of interest (ROI) and the highest value was set as the CT value of the wall. We measured the CT value at any three sites and the average value was calculated. The same method was carried out in non-distended bowel. To measure the CT values in the non-ischemic small bowel, the non-distended small bowel was defined as the small intestine away from the lesion as the measurement point. Thereafter, ROC curve analysis was used to explore the optimal cutoff point. Based on the present cut-off point and the previous study which also evaluated the CT value of intestinal ischemia of SSBO patients [9], reduced wall enhancement was defined as the average CT value of the distended small-bowel wall decreased by 30% compared to that of the non-distended bowel (Fig. 1, Figure S1). There were two clinical outcome categories: patients who underwent laparotomy but had no evidence of ischemia and no resection, and patients who underwent laparotomy with evidence of intestinal ischemia requiring small bowel resection. There were no cases in which intestinal resection was performed for reasons other than ischemia, and in which ischemia progressed and additional bowel resection was performed later. Fig. 1 Measurement method of the CT value of the small bowel wall. 10 mm-white circles were shown as region of interest (ROI). The small-bowel wall was contained in ROI. The highest CT value of ROI was set as the CT value of the wall. Three different sites were measured. A Representative figure of the measurement of distended small bowel. The average value was calculated as the average CT value of distended bowel wall (dCT). B Representative figure of the measurement of non-distended small bowel. The average value was calculated as the average CT value of non-distended bowel wall (nCT). CT, computed tomography. ROI, region of interest

Measurement method of the CT value of the small bowel wall. 10 mm-white circles were shown as region of interest (ROI). The small-bowel wall was contained in ROI. The highest CT value of ROI was set as the CT value of the wall. Three different sites were measured. A Representative figure of the measurement of distended small bowel. The average value was calculated as the average CT value of distended bowel wall

Statistical analysis and the creation of the clinical score to predict intestinal ischemia

Results

ASA-PS American Society of Anesthesiologists physical status, WBC White blood cell, CRP C-reactive protein, CK Creatine kinase, LDH Lactate dehydrogenase, BE Base excess, CT Computed tomography
aMedian (range)

In the DC, 36 patients (36%) required small bowel resection because of ischemia or necrosis (ischemia group), and 64 patients underwent surgery without the need for small bowel resection (no ischemia group). The results of univariate analysis in the DC are shown in Table 2. Higher white blood cell count (WBC $\geq 10,000/\text{L}$), lower base excess (BE $\leq -1.0 \text{ mmol/L}$), and ascites were significant factors in the resection group. Table 2 Predictive factors of intestinal ischemia in development cohort

	No ischemia group (n = 64)	Ischemia group (n = 36)	P	OR (95% CI)
Background Age years ≥ 60 years	46 (72%)	28 (78%)	0.515	
Sex, male/female	24 / 40	13 / 23	0.890	
Past history of surgery	38 (59%)	24 (67%)	0.469	
Duration of symptoms				

before surgery ≥ 24 h / < 24 h 14 (22%) 9 (25%) 0.667 ASA-PS ≥ 253 (82%) 34 (95%) 0.078 Physical findings
 Peritoneal irritation sign 19 (30%) 12 (34%) 0.638 Body temperature ≥ 37.5 °C 6 (10%) 1 (3%) 0.228 Laboratory
 data WBC count $\geq 10,000/L$ 21 (33%) 21 (58%) 0.013 2.86 (1.23—6.66) CRP ≥ 5.0 mg/L 4 (6%) 3 (8%) 0.698 CK
 $\geq 200U/L$ 6 (10%) 3 (9%) 0.867 LDH $\geq 300U/L$ 6 (6%) 5 (14%) 0.228 Arterial blood gas pH ≤ 7.350 (0%) 1
 (3%) 0.153 BE ≤ -1.0 mmol/L 16 (25%) 17 (47%) 0.024 2.68 (1.13—6.38) CT findings Ascites 43 (67%) 32
 (89%) 0.012 3.91 (1.22—12.5) Mesenteric fluid 44 (69%) 29 (81%) 0.194 Reduced bowel wall enhancement 16
 (30%) 24 (80%) < 0.001 9.50 (3.26—27.6) Categorical variables are presented as numbers (%) ASA-PS
 American Society of Anesthesiologists physical status, WBC White blood cell, CRP C-reactive protein, CK
 Creatine kinase, LDH Lactate dehydrogenase, BE Base excess, CT Computed tomography

Predictive factors of intestinal ischemia in development cohort

Categorical variables are presented as numbers (%)

ASA-PS American Society of Anesthesiologists physical status, WBC White blood cell, CRP C-reactive
 protein, CK Creatine kinase, LDH Lactate dehydrogenase, BE Base excess, CT Computed tomography

Eighty-four people underwent contrast-enhanced CT, and reduced bowel enhancement were the
 significant predictive factor for intestinal ischemia. There was no significant difference between the two
 groups regarding the duration of symptoms, peritoneal irritation sign, body temperature, CRP level, or the
 presence of mesenteric fluid.

Development and validation of a scoring system for predicting intestinal ischemia

On the basis of the results of univariate analysis, clinical scores to predict intestinal ischemia requiring
 bowel resection were structured. The ischemia prediction score (IsPS) comprised 1 point each for WBC $\geq$
 $10,000/L$, BE ≤ -1.0 mmol/L, the presence ascites, and 2 points for reduced bowel enhancement (Table 3).
 The simple ischemia prediction score (s-IsPS) includes WBC, BE, and ascites. The modified ischemia
 prediction score (m-IsPS) includes WBC, BE, ascites and contrast-enhanced effect of intestinal wall
 evaluated by enhanced CT. Cut-off values of s-IsPS and m-IsPS were set based on ROC curve analysis
 (Figure S2). The optimal cut-off value was 2 for s-IsPS and 3 for m-IsPS. Table 3 The ischemia prediction
 score (IsPS) Variable Score points White blood cell $\geq 10,000/L$ (WBC) 1 Base excess ≤ -1.0 mmol/L
 (BE) 1 Ascites 1 Reduced bowel wall enhancement 2 [Simple IsPS (s-IsPS)] = WBC + BE + Ascites [Modified IsPS
 (m-isPS)] = WBC + BE + Ascites + Reduced bowel wall enhancement

The ischemia prediction score (IsPS)

[Simple IsPS (s-IsPS)] = WBC + BE + Ascites

[Modified IsPS (m-isPS)] = WBC + BE + Ascites + Reduced bowel wall enhancement

The predictive abilities of each factor and IsPS are shown in Table 4. Among each factor, the presence of
 reduced bowel enhancement was the most valuable predictive factor (sensitivity 70.3%, specificity 80.0%).
 The AUC of s-IsPS was 0.716 and m-IsPS was 0.838, and they were higher than the AUC of each factor. Table
 4 The predictive value of each factor and clinical score in development cohort Sensitivity Specificity AUC
 White blood cell $\geq 10,000/L$ 58.3% 67.2% 0.628 Base excess ≤ -1.0 mmol/L 47.2% 75.0% 0.611 CT:
 Ascites 88.9% 32.8% 0.601 CT: Reduced contrast enhancement 70.3% 80.0% 0.752 s-IsPS $\geq$
 2 69.4% 65.4% 0.716 m-IsPS ≥ 3 86.7% 76.0% 0.838 AUC Area under curve, CT Computed tomography, s-IsPS
 Simple ischemia prediction score, m-IsPS Modified ischemia prediction score

The predictive value of each factor and clinical score in development cohort

AUC Area under curve, CT Computed tomography, s-IsPS Simple ischemia prediction score, m-IsPS
 Modified ischemia prediction score

The sensitivity, specificity, and AUC of s-IsPS were 80.0%, 82.3%, and 0.812, respectively in VC. The
 sensitivity, specificity, and AUC of m-IsPS were 80.0%, 88.2%, and 0.841, respectively in VC.

Discussion

In this study, we examined the risk factors of intestinal ischemia in SSBO patients requiring emergency laparotomy and developed clinical scoring system to predict intestinal ischemia. In summary, univariate analysis showed that high WBC, low BE, ascites and reduced bowel enhancement were significantly associated with bowel resection. Furthermore, clinical scores comprising each predictive factor had higher accuracy to predict intestinal ischemia.

Previous studies have also reported the risk factors (WBC, BE, ascites and reduced bowel enhancement) identified in the present study. The lactate level, BE, or creatinine kinase might reflect anoxic damage of small bowel [10–12], and WBC count or procalcitonin level might reflect inflammatory response [13]. In reports on imaging, ascites was identified as an independent risk factor for bowel resection [5, 14]. Ascites can be clearly observed on nonenhanced CT and it is comparatively easy to interpret its presence. Among them, bloody ascites is often observed in SSBO, and it has been reported that the presence of red blood cells in the peritoneal fluid increases according to the degree of strangulation [15]. In the present study, bloody ascites was also found to be associated with small bowel resection [OR: 3.89 (1.47–10.9), *p* values 0.005], but the OR results were similar to those for ascites overall. Therefore, we used ascites (bloody and non-bloody) as the clinical utility variable. Meanwhile, reduced bowel enhancement was reported as the most specific diagnostic value to predict surgical ischemia in SSBO [16]. This CT finding was considered the result of blockage of the bowel wall arteriovenous circulation. The CT sign of small bowel wall thickness was also reported as valuable for diagnosing ischemia [17], but we did not use this sign considering its difficulty in their assessment and cut-off. Although the presence of peritoneal irritation sign is considered important in the diagnosis of small bowel obstruction [3, 4, 18], we did not observe a significant association between the peritoneal irritation sign and the need for bowel resection. That may be because determining the presence of the peritoneal irritation sign is relatively subjective, and the decision would vary among clinicians.

We developed a predictive scoring system that may be useful for the rapid and accurate diagnosis of intestinal ischemia in SSBO by combining these parameters. The s-IsPS showed similar prediction accuracy with an AUC of 0.716 compared to the previous discriminant formula with an AUC of 0.735 that included ascites, peritoneal irritation sign and lactate [14]. Thereafter, the m-IsPS was structured by including evaluation of contrasted CT with objective criteria, and the diagnostic accuracy was elevated to the AUC of 0.838. Several studies have also included the bowel wall enhancement in the clinical score and the AUC exceeded 0.80 [5, 19]. Although the bowel wall attenuation played a pivotal role in their clinical scores, its radiological sign was based on the assessment of an experienced radiologist, and thus it may be difficult for non-specialized clinicians to use them in actual clinical settings. Furthermore, the more the increase in their clinical parameter or total score, the more complicated is the calculation and use in clinical practice. The present score consists of 4 variables and the total scores range from 0 to 5, striking a good balance between accuracy and usability.

This study has its strength and clinical implications. First, since some patients with asthma, kidney injury, or hyperthyroidism are contraindicated to enhanced CT [20, 21], we separately developed the prediction score with and without enhanced CT. Simple predictive score without enhanced CT may be useful for clinicians to predict the probability of intestinal ischemia, whereas a contrasted CT examination is needed to allow a more accurate assessment of the severity of SSBO. Second, unlike the previous study, we assessed the radiological sign of reduced bowel enhancement with objective criteria, which enables a wide variety of clinicians to interpret bowel-wall attenuation on CT. Third, these models have been validated in the validation cohort, and they may apply to similar settings. This study had several limitations. First, potential biases exist because of the retrospective study design, and the study population was relatively small. Second, we did not include patients who received conservative treatment without emergency laparotomy.

In conclusion, white blood cell (WBC) count, base excess (BE), ascites and reduced bowel wall enhancement were the significant predictive factors for bowel resection in SSBO patients. The clinical score consisting of each parameter allowed early and accurate identification of intestinal ischemia in SSBO.

Supplementary Information

Additional file 1: Figure S1. Receiver operating characteristic (ROC) for predicting small bowel resection with the CT value of ischemic and no-ischemic intestinal wall. Figure S2. Receiver operating characteristic (ROC).

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Authors' contributions

SM contributed to interpretation of the data and drafted the manuscript. SF conceived and designed this study; contributed to acquisition, analysis, and interpretation of the data; and was responsible for drafting, editing, and submission of the manuscript. SM and SF played a significant role in the analysis of the data and helped to draft the manuscript. TT, KY, MH, KN, TH and YO contributed to the data interpretation and critically reviewed the manuscript. KD exerted a major impact on the interpretation of the data and critical appraisal of the manuscript. The author(s) read and approved the final manuscript.

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Availability of data and materials

The data that support the findings of this study are available on request from the corresponding author.

Declarations

Ethics approval and consent to participate

The study followed the Declaration of Helsinki and was approved by the Institutional Review Board at Minoh City Hospital. Informed consent was obtained from all the study participants.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

References

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